

VITAL Workshop Tutorials

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March 2025

1 Software Setup

For setting up software, please follow the instructions detailed here: https://finbarargus.github.io/circulatory_autogen/getting-started/

2 0D Bond Graph Blood Flow Tutorial

Follow the demonstration on the whiteboard. Then complete the following tasks:

1. Write out the bond-graph diagram for a pressure inlet, pressure outlet vessel.
2. Derive the system of equations for the vessel. Knowns:
 - $C = 1 \cdot 10^{-8} \text{ m}^3 \cdot \text{Pa}^{-1}$
 - $R_1 = 1 \cdot 10^5 \text{ Pa} \cdot \text{s} \cdot \text{m}^{-3}$
 - $R_2 = 2 \cdot 10^5 \text{ Pa} \cdot \text{s} \cdot \text{m}^{-3}$
 - $I_1 = 10 \text{ Pa} \cdot \text{s}^2 \cdot \text{m}^{-3}$
 - $I_2 = 10 \text{ Pa} \cdot \text{s}^2 \cdot \text{m}^{-3}$
 - $P_{in} = 12 + 2 \sin(2\pi t) \text{ kPa}$
 - $P_{out} = 1 \text{ kPa}$
3. Implement the equations in OpenCOR. Plot the pressure at the centre of the vessel and the flow at the inlet.
 -
4. Extra: Swap the tube law with the following:

$$P = P_0 + K \left(\left(\frac{q}{q_0} \right)^m - \left(\frac{q}{q_0} \right)^n \right) + P_{ext} \quad (1)$$

- Discuss whether (P_0, q_0) should be calibrated from data, or $P_0 = 0$ and $q_0 = q_{us}$ should be used. Simulate the differences in OpenCOR.

3 Circulatory System Generation Tutorial

1. Extract the `simple_physiological_nonlinear_visco.zip` file to a directory of your choice. We will call this dir `{CA_user_dir}` and the circulatory Autogen directory `{CA_dir}`.
 - The `CA_user_dir/resources` directory contains input files that define the modules that are used in your models and their parameters (and later the information for parameter identification).
 - The `CA_dir` contains src files for Circulatory Autogen and configuration files for modules (defining the mathematics and the allowable connections for the modules).
2. Open `{CA_dir}/user_run_files/user_inputs.yaml` and set

```
user_inputs_path_override: {CA_user_dir}/simple_physiological_user_inputs.yaml
```

Important: For Windows, make sure you use “C:” at the start of the directory and backslash instead of forward-slash.

This lets CA know where to look for config files for the model you want to generate and eventually calibrate. The files that are needed to generate this circulatory system model are:

- `resources/simple_physiological_nonlinear_visco_vessel_array.csv` : This file defines the modules that are used and how they are connected together. Figure 1 shows the model that we will first generate.

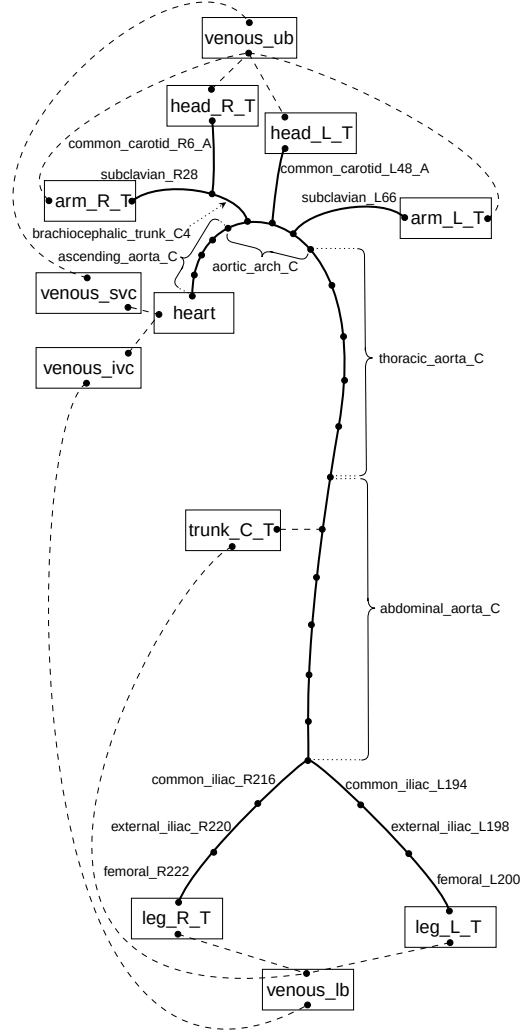
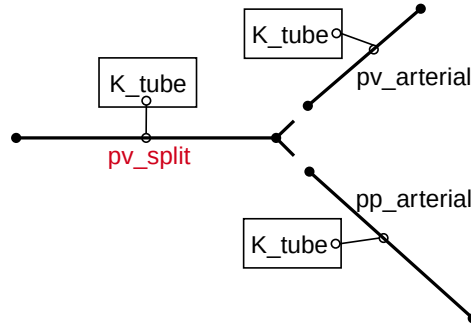


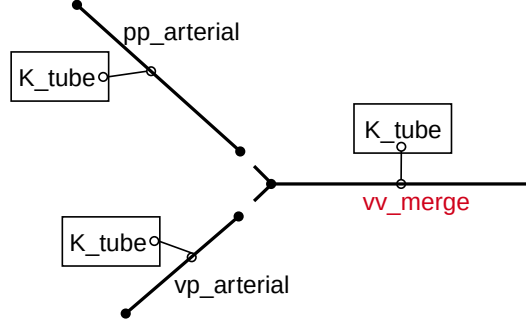
Figure 1: Schematic of the simple physiological CVS model.

- `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_parameters.csv`: This file defines the constants and boundary conditions needed to simulate the model.
- `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_params_for_id.csv`: This file defines the parameters that will be identified in the calibration steps (not modified in this tutorial but explained in https://finbarargus.github.io/circulatory_autogen/parameter-identification/).
- `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_obs_data.json`: This file defines the experimental/clinical protocols and the ground truth values that you want your outputs to replicate (also not modified in this tutorial).
- `{Ca_user_dir}/simple_physiological_user_inputs.yaml`: This is the main configuration file for your model, it specifies the location of your input files, the solver info, the parameter identification algorithm info, and more (you shouldn't need to edit it for this tutorial).

3. Navigate to `{CA_dir}/user_run_files/` and execute the following command to generate the model:



(a) Example of vessel modules to use to define a **bifurcation**: “split_junction” parent vessel dividing into “pv” and “pp” type daughter vessels.



(b) Example of vessel modules to use to define a **merge junction**: “vp” and “pp” type vessels joining into a “merge_junction” vessel.

Figure 3: Schematic of some vessel modules to model junctions. Each vessel module is also coupled to a “K_tube” module to compute its material properties (e.g., the stiffness parameter K in the nonlinear elastic tube law (1)).

- You will need to use a combination of “arterial”, “split_junction” (1-in-2-out) and “merge_junction” (2-in-1-out) to model the superficial palmar arch loop (see Figure 3).
TODO Fig
- BC types must be compatible, an output v (e.g., `pv_nonlinear_visco`) must connect to an inlet p (e.g., `pp_nonlinear_visco` or `pv_nonlinear_visco`) and vice versa. Use `nonlinear_visco` for all.
- Currently, we enforce that the split/merge side of a vessel needs to have a flow boundary condition, therefore, `merge_junction` must be `vv` or `vp` type and `split_junction` must be `vv` or `pv` type
- Re-calculate estimates of terminal resistance R_T for $\{ \text{arm_L_T}, \text{digital_L_1_T}, \text{digital_L_2_T} \}$ according to the following formula

$$R_{T_j} = \frac{P_{art} - P_{ven}}{CO * BFF_j}. \quad (2)$$

Assume that the desired cardiac output in the model is $CO = 100 \cdot 10^{-6} \text{ m}^3/\text{s}$, estimated mean arterial and venous pressures are $P_{art} = 12000 \text{ Pa}$ and $P_{ven} = 666 \text{ Pa}$, respectively, and the desired fractional flows to the three terminals are:

Terminal	BFF (%)	C_T ($\text{m}^3 \cdot \text{Pa}^{-1}$)
arm_L_T	1.75%	$1 \cdot 10^{-10}$
digital_L_1_T	0.875%	$1 \cdot 10^{-11}$
digital_L_2_T	0.875%	$1 \cdot 10^{-11}$

Table 1

Set the terminal compliance to be small for now (see Table 1). These parameters could be further calibrated to PPG data.

7. Again, navigate to `{CA_dir}/user_run_files/` and do the following command to generate the model with your new arm circulation connected.

```
./run_autogeneration.sh
```

8. Open the generated model and plot the digital_L2 pressure.
9. Extra: Navigate to {CA_dir}/user_run_files/ and do the following command to run a parameter identification algorithm on the generated model:

```
./run_param_id.sh {number_of_cores}
```

this will run the optimisation algorithm defined in

```
{CA_user_dir}/simple_physiological_user_inputs.yaml
```

to calibrate the parameters defined in

```
{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_params_for_id.csv
```

to the ground truth values defined in

```
{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_obs_data.json
```

You can the run the following command to plot and compare the model outputs with the ground truth observables:

```
./plot_param_id.sh
```

All identified parameter information and plots are stored in

```
{Ca_user_dir}/param_id_output/
```

For more information on using Circulatory Autogen for parameter identification, see https://finbarargus.github.io/circulatory_autogen/parameter-identification/

4 Cell Modelling With Bond Graph Tutorial

1. Extract the `smc_hernandez.zip` file to a directory of your choice. We will now call this dir {CA_user_dir}" and the Circulatory Autogen directory {CA_dir}.
 - Reminder, the `CA_user_dir` contains input files that define the modules that are used in your models and their parameters (and later the information for parameter identification).
2. Open {CA_dir}/user_run_files/user_inputs.yaml and set

```
user_inputs_path_override: {CA_user_dir}/smc_hernandez_user_inputs.yaml
```

This model is for the smc by Hernandez Hernandez, which we have replicated in CellML and ensured it satisfies BG principles:

TODO: Hernandez' diagram for the SMC model.

3. Navigate to {CA_dir}/user_run_files/ and do the following command to generate the model.

```
./run_autogeneration.sh
```

The model will be generated in

`{CA_user_dir}/generated_models/smc_hernandez`

4. Open the generated model in OpenCOR (`smc_hernandez.cellml`) and plot the membrane potential, V .
5. Now we are going to include a piezo ion channel to allow us to model the response to stretch of a vessel. Draw the BG diagram for the piezo channel (stretch can be an external input) and how it connects to the rest of the model.

TODO the start of the Piezo BG diagram

6. Derive the conservation of current equations for the piezo channel.
7. Open the file `{CA_dir}/module_config_user/user_cell_modules.cellml` in OpenCOR and edit the module named `smc_hernandez_type` to include the derived piezo channel equations and the following empirical model for the piezo channel:

TODOPiezochannelequations

Make sure the piezo current is added to the differential equation for voltage (from the conservation of current derivation).

Use the following parameter values for the Piezo channel:

TODOPiezoparameters

8.

5 Blood Flow - Cellular Coupling Tutorial

Now we will couple the SMC model that we developed in the tutorial from section 4 with a vessel module within our circulatory system model, so that the vessel incorporates myogenic control effects.

1. Open the file `{CA_dir}/module_config_user/vessel_properties_modules.cellml` in OpenCOR. Here, we will develop a module that calculates a changing stiffness, K_{tube} , dependent on the SMC Ca concentration.
 - IMPORTANT NOTE: This simple module will not take into account myofiber mechanics, vessel geometry effects, etc. It is used to demonstrate how cell models can be coupled to vessel models in this framework. The development of physiologically accurate control of vessel constriction from smc dynamics will be done within VITAL.
2. Add the ports to `K_tube.calc` and `smc` model so they can couple together.

TODO give guide in the config file where to do this.